Course program and reading list

Semester 2 Year 2025

School: Adelson School of Entrepreneurship B.A

Introduction to Data Science

Lecturers:

Mr. Sahar Millis sahar.millis@post.runi.ac.il

Tutors:

Ms. Ilana Sivan ilana.sivan@post.runi.ac.il

Teaching Assistants:

Mr. David German david.german@post.runi.ac.il

Course No.:	Course Type :	Weekly Hours :	Credit:
2686	Lecture	3	4

Course Requirements :	Group Code :	Language:
Final Paper	252268609	English

Course Description

This course equips you with practical data science skills to tackle challenges in business, economics, and entrepreneurship. You'll build a robust foundation in data analysis and machine learning - using Python, AI technologies, and industry-standard tools. Through hands-on projects, you'll develop predictive models, perform customer segmentation, and make data-informed decisions using key techniques such as regression and clustering. Whether you're a data novice or have some experience, you'll graduate with a powerful toolkit applicable to your studies or entrepreneurial pursuits.

Course Goals

The course will cover the following topics.

Unit #	Title	Topics
1	Introduction to Data Science	What is data science, statistics, ML, and DL? Why is it important for business? The data science lifecycle
2	Data & EDA & Visualization	Sources of data, Data formats, Data cleaning techniques Descriptive statistics, Identifying patterns, and trends Visualizations in communication, plots, and graphs
3	Intro to Machine Learning and Classification	What is ML? What is Classification? Use cases: Fraud detection, customer segmentation Metrics: Accuracy, Precision, Recall, F1-score
4	Regression	What is Regression? Use cases: Predicting stock/property prices Metrics: MSE, RMSE, R-squared
5	Clustering	What is Clustering? Use cases: Customer segmentation, market research, product categorization Metrics: Silhouette score, elbow method
6	Predictive Analytics	What is Predictive Analytics? Use cases: Sales forecasting, customer churn prediction Metrics: MAE, RMSE, cross-validation
7	Recommendation Systems	What are recommendation systems? Popularity, trending, Collaborative, Content-based Use cases: Product recommendations, cross-selling Metrics: Precision, Recall, MAP, cold-start
8	Intro to Deep Learning	What is DL? ML vs DL? What are Features and Embeddings? Use cases: Image recognition, chatbots, predictive text Metrics: confusion matrix
9	Textual Data (NLP)	What is NLP? Use cases: Sentiment analysis, topic modeling, chatbots Metrics: Precision, Recall, F1-score, similarity
10	Visual Data (Images)	What is Image Analysis? Use cases: Product categorization, defect detection, facial recognition Metrics: Accuracy, precision, recall

11	Time & Behavioral Data	What are time series and behavioral data? Use cases: Sales forecasting, user behavior tracking Metrics: MSE, RMSE, customer journey metrics
12	Course Summary & Projects	Recap Final projects Feedback & Reflection

* Some adjustments may occur.
** Guest lecture dates will be published during the course.
** Topics covered in advanced classes may change based on class advancement.

Grading

60% Assignments (4 in total)

40% Final Project

Lecturer Office Hours

Before and after class.

Additionally, upon request - online during the week.

Tutor Office Hours

TBD

Teaching Assistant

TBD

Additional Notes

Course attendance is aligned with the Entrepreneurship School's guidelines.

About the lecturer.

Reading List

Online:

1. python.org
2. scipy.org
3. scikit-learn.org
4. huggingface.com

BI Tools:

1. tableau.com
2. powerbi.com

Generative Tools:

1. claude.ai
2. chatgpt.ai
3. perplexity.ai
4. gemini.google.com
5. groq.com (free API for dev)

Books:

1. "Storytelling with Data" by Cole Nussbaumer Knaflic
2. "Data Visualization: A Practical Introduction" by Kieran Healy
3. "The Hundred-Page Machine Learning Book" by Andriy Burkov
4. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
5. "Introduction to Time Series and Forecasting" by Peter J. Brockwell and Richard A. Davis

Online Courses (free)**:**

1. https://www.edx.org/learn/data-visualization/harvard-university-data-science-visualization
2. https://www.edx.org/learn/machine-learning/harvard-university-data-science-machine-learning
3. https://www.edx.org/learn/data-science/harvard-university-introduction-to-data-science-with-python
4. https://www.edx.org/learn/machine-learning/harvard-university-machine-learning-and-ai-with-python
5. https://www.edx.org/learn/artificial-intelligence/harvard-university-cs50-s-introduction-to-artificial-intelligence-with-python